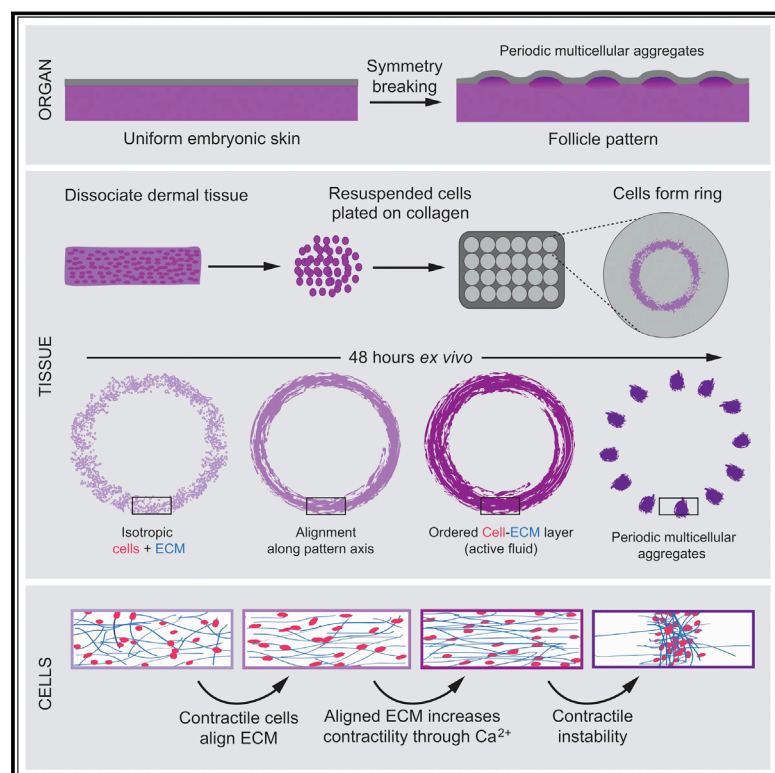


Reciprocal cell-ECM dynamics generate supracellular fluidity underlying spontaneous follicle patterning

Graphical abstract



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In brief

Contractility-dependent mechanical interplay between cells and the extracellular matrix (ECM) generates an ordered cell-ECM layer with fluid instability features, leading to spontaneous follicle patterning in an *ex vivo* tissue reconstitution model.

Highlights

- An *ex vivo* assay reconstitutes the process of embryonic skin symmetry breaking
- Mesenchymal mechanics are sufficient to spontaneously generate morphological pattern
- Contractility-dependent cell-ECM interplay creates supracellular structure
- Theory and experiment show that the cell-ECM layer behaves as an active contractile fluid



Article

Reciprocal cell-ECM dynamics generate supracellular fluidity underlying spontaneous follicle patterning

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SUMMARY

During vertebrate embryogenesis, cell collectives engage in coordinated behavior to form tissue structures of increasing complexity. In the avian skin, assembly into follicles depends on intrinsic mechanical forces of the dermis, but how cell mechanics initiate pattern formation is not known. Here, we reconstitute the initiation of follicle patterning *ex vivo* using only freshly dissociated avian dermal cells and collagen. We find that contractile cells physically rearrange the extracellular matrix (ECM) and that ECM rearrangement further aligns cells. This exchange transforms a mechanically unlinked collective of dermal cells into a continuum, with coherent, long-range order. Combining theory with experiment, we show that this ordered cell-ECM layer behaves as an active contractile fluid that spontaneously forms regular patterns. Our study illustrates a role for mesenchymal dynamics in generating cell-level ordering and tissue-level patterning through a fluid instability—processes that may be at play across morphological symmetry-breaking contexts.

INTRODUCTION

The skin is the largest organ in the body, requiring long-range coordination to achieve proper functional architecture. The most decisive morphological event in the skin occurs during embryogenesis, when a flat tissue bilayer of dermis and epithelium transitions into one studded with an array of follicle primordia. Each primordium emerges as a multicellular aggregate with an activated follicle gene expression program. This architecture serves as the morphological template for the rest of skin development and persists into the adult body. Decades of research has been conducted under the presumption that for such a robust pattern to form, patterns of gene expression must emerge prior to the onset of new forms and therefore regulate the initiation of follicle gene expression as well as structure (Jung et al., 1998; Mandler and Neubüser, 2004; Painter et al., 2018). Recent work that compared the dynamics of gene expression and morphological changes in the avian skin showed that the expression of localized signaling molecules does not precede morphological changes (Shyer et al., 2017). Instead, mechanical self-organization of the dermis is sufficient to initiate a periodic pattern and also initiate follicle gene expression through β -catenin-dependent mechanotransduction.

While the mechanism of how the dermal layer intrinsically generates tissue scale pattern remains unknown, it has been shown that the determining forces are dependent on the contractile behavior of the mesenchymal cells within the dermis (Shyer et al., 2017). Contractile behavior in mesenchymal cells has long been appreciated in studies of cultured fibroblasts (Bell et al., 1979; Harris et al., 1981; Baker et al., 2015), and the skin presents an opportunity to articulate how such individual cellular forces coalesce in a collective to drive morphogenesis. Such a mechanism, whereby multicellular self-organization acts as the regulatory step for initiating both tissue-scale morphology as well as gene expression, has received far less attention in studies of vertebrate organogenesis than models based on molecular pre-patterning.

Importantly, unlike molecular mechanisms, which exhaustively exploit a single scale (Peter and Davidson, 2017; Raspopovic et al., 2014), mechanisms articulating multicellular self-organization provide a natural opportunity for coverage across scales, connecting events from the molecule to the organ. Namely, collective cell mechanics can be tied to the state of the cellular components and underlying molecular regulators thereof. Further, organ and tissue structure can be understood as an outcome of spontaneous transitions in the cellular

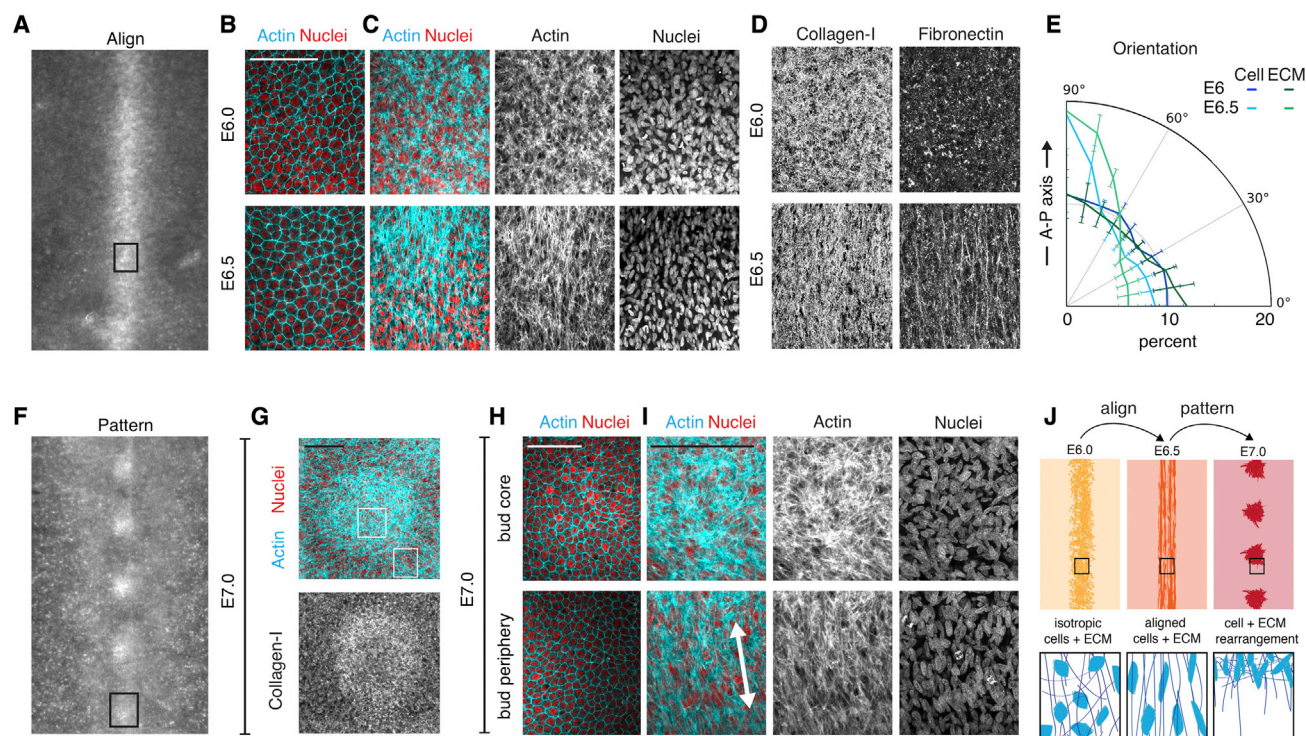


Figure 1. Dermal cells display supracellular order preceding feather follicle emergence

(A) Brightfield image of day 6.5 embryo back skin.

(B and C) Maximum intensity projections of F-actin (phalloidin) and nuclei (DAPI) in the (B) epidermis and (C) dermis. Scale bars, 100 μ m.

(D) Maximum intensity projections of collagen-I and fibronectin in days 6 and 6.5 dermis decellularized with SDS.

(E) Plot showing orientation of cells (F-actin) and ECM (fibronectin) from days 6 to 6.5. Data are represented as mean $\pm$ SD.

(F) Brightfield image of day 7.0 embryo back skin.

(G) Maximum intensity projections of a single feather follicle primordium stained for nuclei, F-actin, and collagen I. Scale bars, 100 μ m.

(H and I) High-resolution maximum intensity projections of the primordium core and periphery in the (H) epidermis and (I) dermis showing F-actin and nuclei. Scale bars, 100 μ m. White arrow is radially oriented to the bud core.

(J) Schematic of the timing of alignment and patterning *in vivo*.

See also Figure S1.

collective. The lack of multiscale mechanisms in development and morphogenesis has recently been highlighted as a fundamental challenge in the field (Lenne et al., 2021). The skin, with its simplicity of geometry (a flat sheet) and pattern (a grid), serves as an ideal model system to confront the complexity of a multiscale mechanism of morphogenesis.

Here, we gain insight into how multicellular self-organization of the dermis leads to ordered tissue structure and determine the molecular and cellular interactions underlying such a spontaneous tissue transition. We do so through the development of an *ex vivo* tissue reconstitution, where we recapitulate the ordered follicle pattern in a collective of dermal fibroblasts without any potential organizing molecular cues from neighboring tissues. Experimental and theoretical analysis of the relationship between individual cell features and tissue-scale properties serve to elucidate fundamental principles underlying active soft matter, a still poorly understood class of materials in which the individual constituents within a material consume energy and dynamically adapt to their extrinsic environment (van der Gucht, 2018). In this context, the long-range structural properties of a supracellular material provide a conceptual framework for un-

derstanding how cells robustly generate long-range morphological order across an organ.

RESULTS

Characterizing transitions in skin architecture at the multicellular scale

To address the question of how symmetry is broken in the skin, we investigated which multicellular features set the conditions for initiation of a follicle pattern. We took advantage of a critical feature of chicken skin patterning, which is that it initiates as a single row of follicles that propagates laterally across a field (Figures 1A and 1F). The progressive nature of pattern formation simplifies the consideration of organ symmetry breaking to a single dimension. Therefore, to address the question of periodic patterning in the skin, we sought to explain the transition of a uniform band of cells (Figure 1A) into a row of regularly spaced multicellular aggregates (Figure 1F).

For decades, the cell-extracellular matrix (ECM) architecture has been hypothesized to be a nexus for morphological changes (Ettinger and Doljanski, 1992). We therefore characterized cell

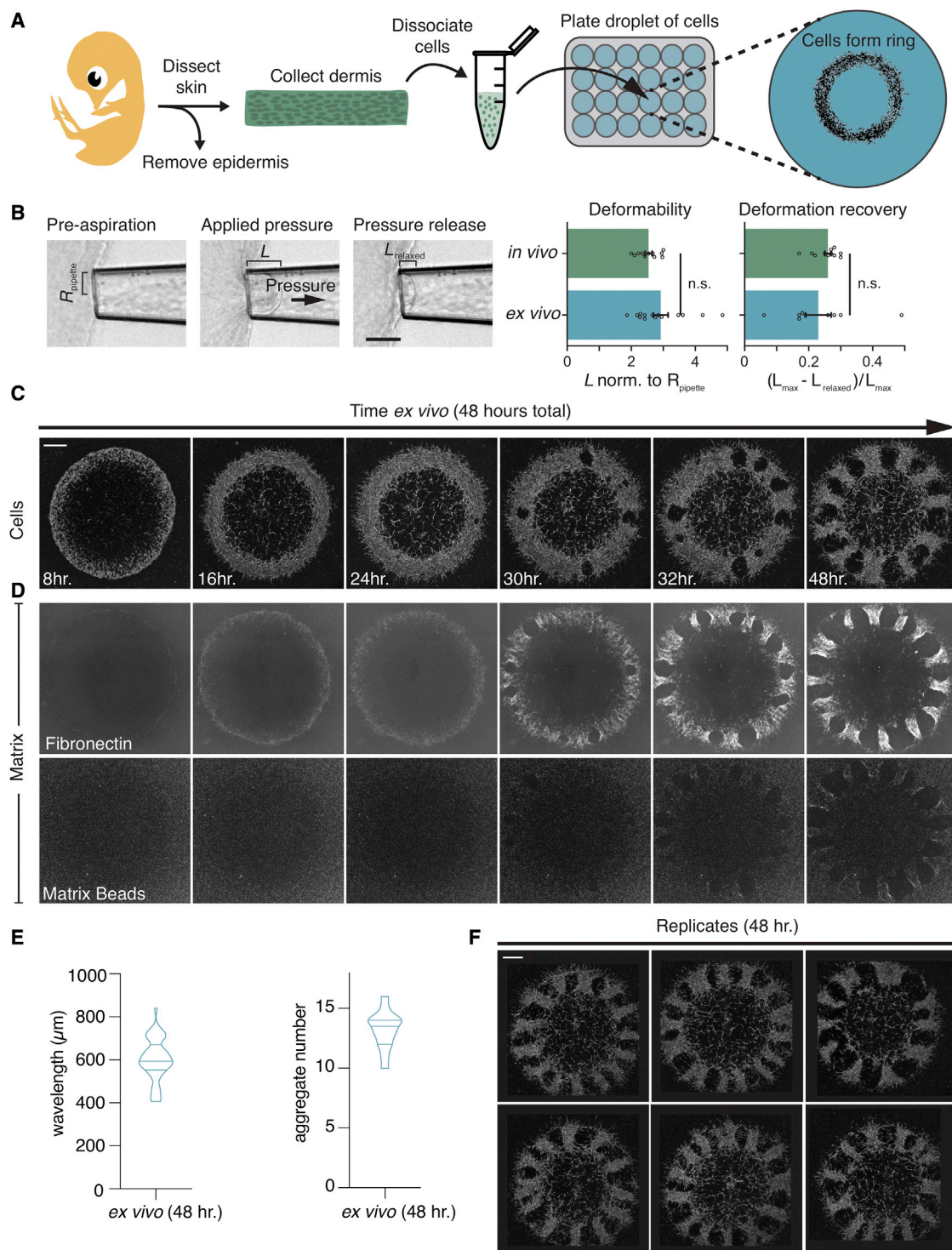


Figure 2. A novel assay reconstitutes multicellular behaviors and tissue patterning *ex vivo*

(A) Schematic representation of *ex vivo* reconstitution assay showing removal of skin from embryo, isolation and dissociation of dermis, and resuspension on collagen-I gel.

(B) Micropipette aspiration of *in vivo* tissue prior to follicle patterning and of the *ex vivo* collagen-I gel. Plots show similarities between *in vivo* and *ex vivo* in terms of deformability (quantified as length of sample extension L into the pipette, normalized to pipette radius R_{pipette}) and deformation recovery (fraction of sample extension that reverses after aspiration pressure is released, see [STAR Methods](#)). Scale bars, 25 μm .

(C) Transformed brightfield images of dermal fibroblasts, plated in an annular geometry, over a 48-h period *ex vivo*. Scale bars, 500 μm .

(legend continued on next page)

and ECM architecture at the collective cell scale. At embryonic day 6 (E6.0), the cells and ECM within the pattern-forming band of dermis that runs anterior-posterior (A-P) through the center of the back are isotopically organized. Over the course of the next 12 h (E6.0–E6.5), F-actin and nuclei acquire a clear orientation along the A-P axis of the body (Figures 1C and 1E). Notably, F-actin anatomy and nuclear structure remain similar in the epidermis from E6.0 to E6.5 (Figure 1B). This supports previous studies discovering that mesoscale structural changes within the dermis precede force transmission to the epidermis (Shyer et al., 2017). To complement cellular architecture, we characterized ECM changes through the staining of collagen-I and fibronectin, two major components of the skin ECM. Examination of the fibronectin meshwork reveals a greater degree of orientation along the A-P axis at E6.5 as compared with E6.0 (Figures 1D and 1E).

Following cell-ECM alignment, over the next 12 h (from E6.5 to E7.0) the aligned band distributes into regularly spaced aggregates (Figure 1F). We find the presence of continued collective cell alignment radially around the periphery of aggregates but the absence of alignment at the core (Figure 1I). At the same time as this transition in the dermis, the epidermis shows the expected morphological shift, with epithelial cells above a nascent aggregate constricted *en face* (Figure 1H). At E7.0, we note increased fibronectin and collagen I accumulation in nascent feather follicle primordia (Figure 1G), evidencing that the formation of this periodic pattern involves not just the rearrangement of cells but also the ECM.

This characterization reveals a tissue-wide A-P orientation of cells and ECM that precedes the formation of a row of aggregates (Figure 1J). Although heterogeneity in molecular positional information affords the most obvious pattern pre-condition, our results suggest that a uniform transformation across the pattern-forming band lays the foundation. We tested whether such a transformation in the cells and ECM alters tissue-level physical properties. In line with this hypothesis, micropipette aspiration (MPA) reveals that aligned skin deforms less than unaligned skin in response to constant applied force (Figure S1). This finding compelled consideration of how shifts in cell and ECM organization occur and, further, how these shifts endow the tissue with mechanical properties that, when coupled with appropriate tissue geometry, induce the formation of novel tissue-scale structures.

Multicellular reconstitution of a spontaneous morphological process

To investigate how a regularly spaced row of aggregates emerges from an isotropic band of cells, we engineered a novel *ex vivo* platform to recapitulate this transformation outside of the body. Such a platform is necessary to determine whether key cell and tissue transitions are autonomous, self-organized events within the dermis, and also to dissect such a mechanism across biological scales. Freshly harvested embryonic dermal cells

were dissociated into a single-cell suspension and plated in a drop atop a collagen gel. Through a natural consequence of fluid mechanics termed the “coffee ring effect” (Deegan et al., 1997), the cells settle onto the collagen substrate around the circumference of the drop (Figure 2A). Following an hour’s incubation allowing cells to settle and attach, media was added to cover the cells. Geometrically, given the width of the ring of cells is small compared with its radius, the additional curvature can be neglected, and the ring serves as a line with no end (González et al., 2013), thereby removing confounding irregularities at the edge. This ring state then mirrors the band of isotopically organized cells at E6.0 (Figure 1A). The three-dimensional collagen gel substrate with specific density, volume, and gelation conditions is a necessary component to recapitulate the properties of the collagen-rich substrate beneath the dermal cells *in vivo*. Using MPA, we confirmed that the collagen gel used *ex vivo* was comparable in deformability and deformation recovery to the *in vivo* back skin dermis (Figure 2B). Combined with a high-throughput, live-imaging platform, our *ex vivo* platform allows for live-cell and ECM tracking in large numbers of samples.

When tracking cell arrangement over time, we find our *ex vivo* reconstitution recapitulates the spontaneous formation of ordered aggregates observed in the skin (Figures 2C and S2; Video S1). These appear as progressive segregations of both the cells and the ECM (Figures 2C and 2D; Video S2), recapitulating the coupling between the cells and the ECM *in vivo*. We observe regularity of aggregate pattern within a ring as well as across trials (Figures 2E and 2F). These results demonstrate that dermal cells autonomously generate an ordered pattern of aggregates without the requirement of cues from the epidermis, as previously suggested (Ho et al., 2019). Importantly, not all types of fibroblasts tested in this culture system undergo such transitions in our *ex vivo* platform. In particular, a chick embryo fibroblast cell-line fails to properly form a pattern (Figure S2), highlighting the specificity of this patterning forming program and the importance of using fresh primary cells to model an *in vivo* process.

Dermal cells autonomously generate supracellular order through ECM alignment

Our *ex vivo* system, where the active component is solely dermal cells, provides an ideal context to investigate the origin and role of the *in vivo* architecture we uncovered. If this system does not recapitulate the alignment of cells and ECM *ex vivo*, this would suggest that this transition stems from forces or signals external to the dermal layer of the skin, such as the underlying tissue or axial growth of the body. Further, lack of alignment in the context of ordered pattern formation *ex vivo* would imply that this step is dispensable for pattern formation.

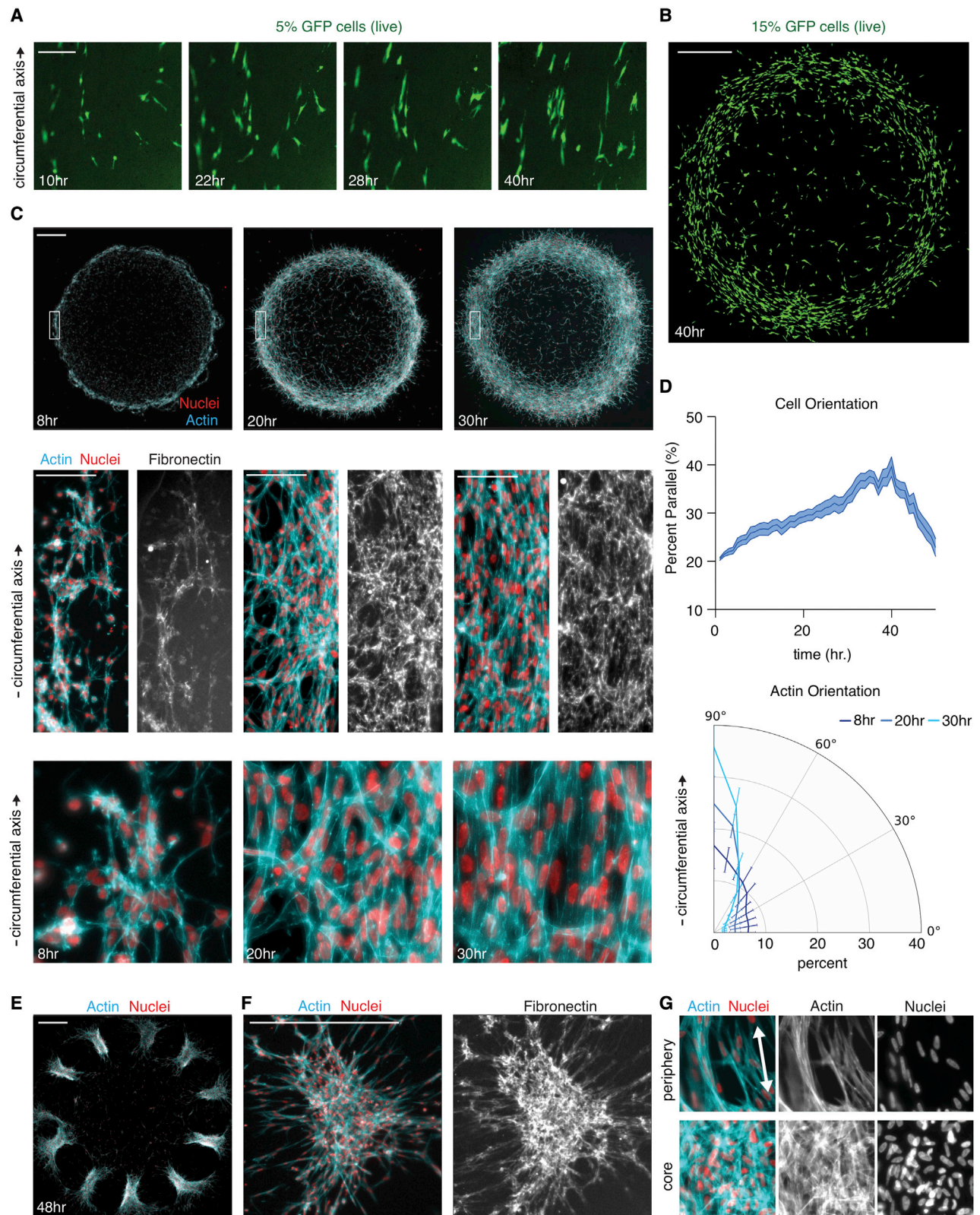
Instead, we find that we recapitulate cell-ECM alignment *ex vivo*. In both live-cell imaging of GFP-labeled cells (Figures 3A, 3B, and 3D; Video S2) as well as fixed samples stained for cell morphology markers (actin and nuclei) and

(D) Fluorescent imaging of conjugated AF555-fibronectin and matrix-embedded GFP beads.

(E) Plots showing the distribution of average wavelength and aggregate number at 48-h post plating *ex vivo* over multiple trials.

(F) Replicates of 48-h time point *ex vivo* showing the regularity and robustness of pattern formation. Scale bars, 500 μ m.

See also Figure S2.



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ECM (fibronectin) (Figures 3C and 3D), we capture progressive alignment of both cells and ECM linearly along the pattern axis over time. These results demonstrate that the cell-ECM alignment observed *in vivo* is self-organized from within the dermis. Further, the fact that the link between this aligned state and aggregation is maintained *ex vivo*, supports the necessity of this conformation in the process of follicle formation. Of note, upon the formation of aggregates *ex vivo*, cells at the core become disordered and those at the periphery appear radially oriented (Figures 3E–3G). Therefore, we also recapitulate *ex vivo* the cell arrangement of the *in vivo* follicle primordia.

We investigated the mechanism responsible for this critical self-organization into an aligned cell-ECM band. When considering what features of the dermis might allow the tissue to align itself, we hypothesized that cellular contractility serves as the force generator. Indeed, we find that when contractility is decreased through myosin II inhibition, nuclei and fibronectin fail to orient (Figures 4A and 4B). Further we inhibited contractility in whole skin explants and find we recapitulate these results in the intact skin (Figure S3). Given the coupling between the ECM and cell rearrangement, we explored the role of the ECM's material properties. We find that when we increase cross-linking of the collagen gel substrate through ribose-mediated glycation (Francis-Sedlak et al., 2009; Mason and Reinhart-King, 2013; Vicens-Zygmunt et al., 2015), alignment is significantly delayed (Figures 4C and 4D). Therefore, this alignment is dependent on particular ECM physical properties. Together, these results support a mechanism whereby contractile cells rearrange the ECM. Such a realignment of the extracellular environment would serve to further align the cells, generating a reciprocal cell-ECM exchange responsible for collective alignment. This tissue-wide alignment represents the coordination of cells and ECM beyond the length scale of an individual cell (i.e., supracellular order).

From a biophysical perspective, we noted that the supracellular alignment of cells and ECM is similar to nematic ordering in liquid crystals. Although initially studied in passive chemicals, more recently, concepts from liquid crystal physics have been applied to multicellular contexts, including the supracellular organization of adult liver and cultured epithelial cells (Morales-Navarrete et al., 2019; Saw et al., 2017). In the context of tissue morphogenesis, we reason that nematic ordering both indicates and confers supracellular coordination, allowing cellular forces to have tissue-level effects.

Ca²⁺ signaling mediates a cell-ECM co-evolution critical for generating supracellular order

We investigated how the establishment of supracellular order sets the stage for transformation to aggregates. In particular, given that ECM alignment and increasing amounts of fibronectin would predict a supracellular structure that would be less likely to separate (Wang et al., 2014; Ahmadzadeh et al., 2017), why does cell-ECM alignment precede separation into discrete units rather than stabilize the field into a linear strip of oriented cells? The spatial and temporal access inherent to the *ex vivo* platform provides the opportunity to dissect how critical cellular and intracellular mechanisms set the conditions for such a tissue-level transformation.

We hypothesized that increases in cell-ECM alignment might stimulate a concomitant increase in cellular contractility due to increasing tension on the cell, as has been demonstrated in other contexts (Ahmadzadeh et al., 2017; Polte et al., 2004). Thus, cell-ECM alignment would trigger a more contractile dermal layer that could overcome the increasing integrity of the ECM to separate into aggregates. To test whether cells shift contractility over time *ex vivo*, we developed two methods to measure traction forces in cells collected from ring cultures at early (6 h) and late (25 h) time points (Figure 4E). In both methods, we extract cells from the ring and replat to assure equivalent cell arrangements (Figure S3) needed for accurate comparison of traction forces at different times. First, we compared the rate of *ex vivo* rearrangement across cells collected at these time points when dissociated and replated in a new *ex vivo* ring culture. We find a difference in contractile behavior between cells collected at 6 h and those collected at 25 h. Specifically, those reseeded from the late time point initiated aggregation almost immediately, whereas those from the early time point took significantly longer, indicative of cells from later *ex vivo* time points being more contractile (Figure 4F). To test this point directly, we also reseeded cells from these two time points in a collective traction force microscopy (TFM) assay, where contractility could be assayed by deformation of a collagen matrix embedded with GFP beads. In line with the above observations, cells isolated from a later time point contracted the collagen gel more strongly as compared with those cells isolated from an earlier time point (Figures 4G, 4H, and S3).

To identify the mechanism underlying progressively increasing contractility, we investigated the role of Ca²⁺-mediated mechanotransduction, which has been shown in other systems to tune contractility in response to extracellular mechanics

Figure 3. An aligned cell-ECM suprastructure forms over time *ex vivo*

- (A) Live imaging of GFP cells *ex vivo*. Scale bars, 100 μ m.
- (B) A single time point image of the entire *ex vivo* platform with ~15% of cells labeled by GFP. Scale bars, 500 μ m.
- (C) Maximum intensity projections of fixed time points stained for nuclei and F-actin at low (top), medium (middle), and high (bottom) resolution. (Top) Scale bars, 500 μ m. (Middle) Scale bars, 100 μ m.
- (D) Plots of cell alignment over time. (Top) Orientation of GFP-labeled cell body over a 50-h period with orientations binned. Percentage of cells parallel to the circumferential axis, meaning those that are 80°–90° to the angle normal to the circumferential axis, as measured using the directionality plug-in in ImageJ. Mean $\pm$ SEM shown. (Bottom) Orientation of F-actin in a field of cells at three time points. 90° is parallel to the circumferential axis. Measurements were taken in 10° bins using the directionality plug-in in ImageJ. Mean $\pm$ SD shown.
- (E and F) Maximum intensity projections of 48-h time point *ex vivo* at low (E) and high (F) magnification, stained for nuclei and actin as well as fibronectin. Scale bars, 500 μ m.
- (G) High-resolution image of one aggregate.
- (H) Maximum intensity projection of the core and periphery of a single aggregate *ex vivo*, stained for nuclei and actin. The white arrow is radially oriented toward the core of the aggregate and is roughly parallel to the circumferential axis.

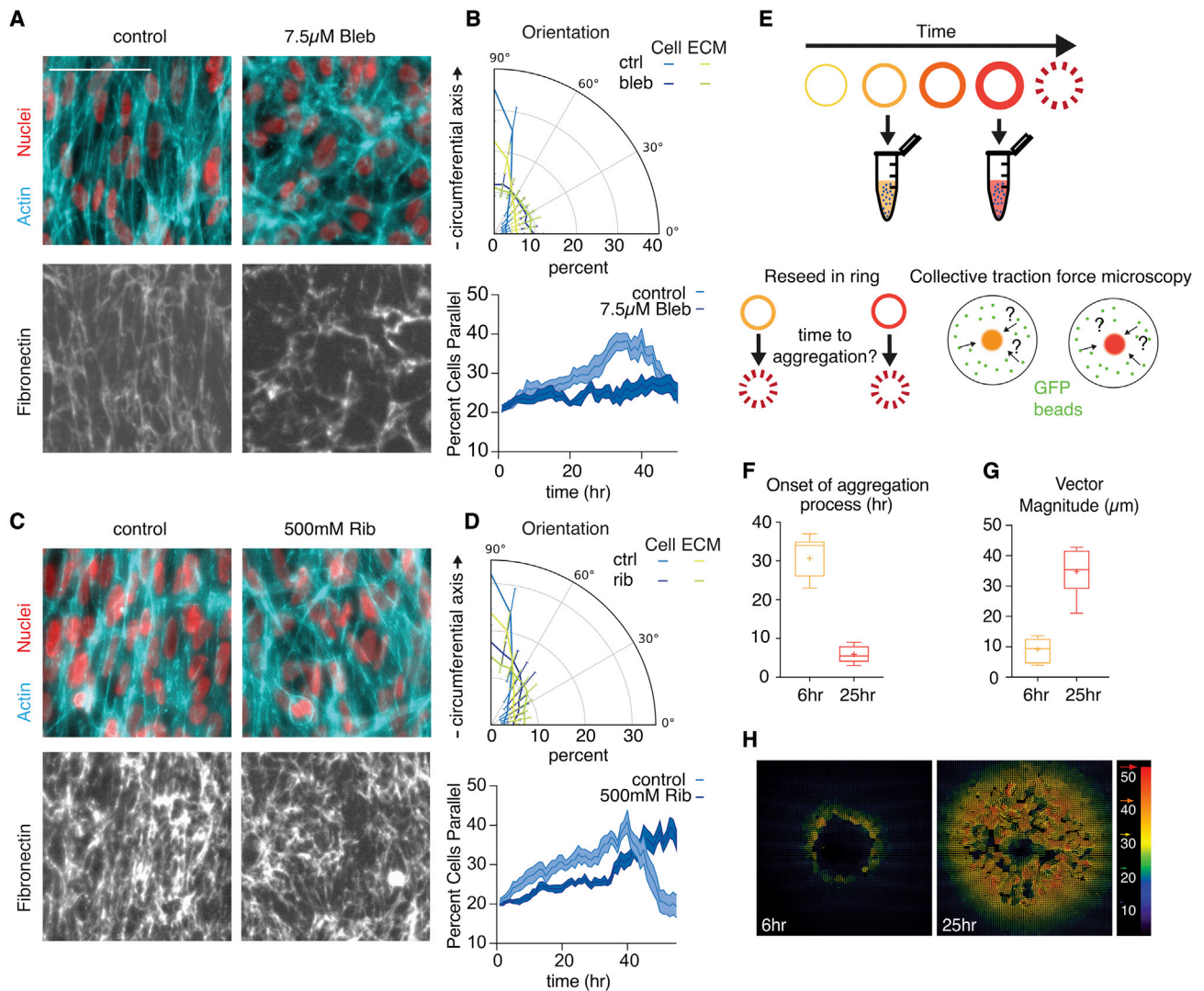


Figure 4. Cellular contractility and ECM properties regulate the formation of supracellular structure

(A) Maximum intensity projection of F-actin, nuclei, and fibronectin in control and 7.5- μ M blebbistatin-treated *ex vivo* cultures. Scale bars, 50 μ m.
 (B) Plots of cell and ECM orientation in control and blebbistatin-treated *ex vivo* cultures. (Top) Radial plot showing cell (F-actin) and ECM (fibronectin) orientation. Mean $\pm$ SD is shown. (Bottom) Live-cell orientation of GFP-labeled cell bodies in control and 7.5- μ M blebbistatin-treated *ex vivo* cultures. Mean $\pm$ SEM is shown.
 (C) Maximum intensity projection of F-actin, nuclei, and fibronectin in control and cross-linked (with 500 mM ribose) *ex vivo* cultures.
 (D) Plots of cell and ECM orientation in control and cross-linked *ex vivo* cultures. (Top) Radial plot showing cell (F-actin) and ECM (fibronectin) orientation. Mean $\pm$ SD is shown. (Bottom) Live-cell orientation of GFP-labeled cell bodies in control and cross-linked *ex vivo* cultures. Mean $\pm$ SEM is shown.
 (E) Passing of cells at early and late time points from the *ex vivo* system into (left) a subsequent *ex vivo* assay and (right) into a bead contractility assay.
 (F) Plot showing the onset of aggregation in hours for cells passaged early (6 h) and late (25 h).
 (G) Plot of the average vector magnitude (μ m) across multiple trials for cells passaged into a bead contractility assay isolated early (6 h) and late (25 h).
 (H) PIV magnitude map of vectors from an individual, representative bead contractility assay performed over a 2-h time interval.
 See also Figure S3.

(Munevar et al., 2004; Matthews et al., 2010; Kim et al., 2014). In line with such a role for Ca^{2+} in our system, we find that cells are positive for fluorescent imaging plate reader (FLIPR) Calcium 6 staining, a marker of Ca^{2+} signaling (Figure 5A; Video S3; Sirenko et al., 2013). Further, from 12 to 32 h, cells on average double the frequency of their calcium oscillations, which is consistent with a role for calcium in increasing contractility over time, *ex vivo* (Figures 5A and 5B; Doyle et al., 2004). To investigate whether

dermal cells respond to changes in ECM deformability by changing Ca^{2+} signaling frequency, we measured Ca^{2+} signaling on increasingly cross-linked collagen gels. Indeed, we find that dermal cells on more cross-linked collagen gels have significantly increased Ca^{2+} signaling frequency as compared with controls (Figure 5C). In line with a role for Ca^{2+} signaling in increasing contractility necessary for the transition to ordered aggregates, we found that inhibiting Ca^{2+} signaling using the

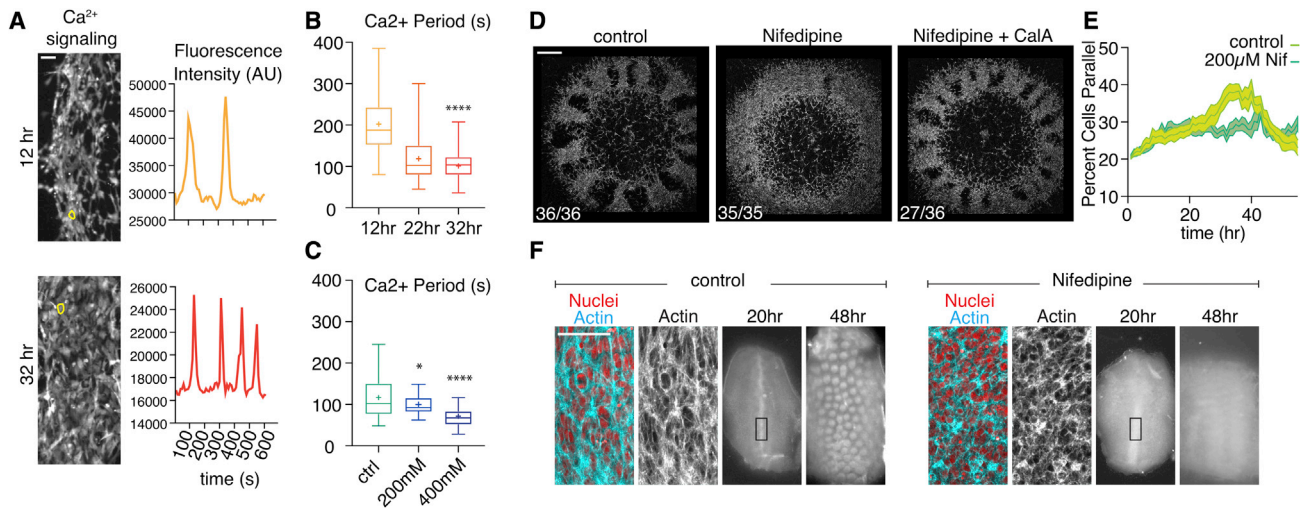


Figure 5. Mechanosensitive calcium signaling tunes patterning in a contractility-dependent manner

(A) Representative images and plots of live calcium signaling dye (FLIPR Calcium 6) showing peaks of calcium signaling activity. Scale bars, 50 μ m. (B) Plot of average period of calcium signaling across trials at three time points *ex vivo*. Statistical comparison shown is between the 12- and 32-h time points. (C) Plot of average period of calcium signaling at 22 h post plating on collagen-I gels cross-linked with ribose treatment. Statistical comparisons shown are between cross-linked gels and control. (D) Transformed brightfield images of control, 100- μ M nifedipine, and 100- μ M nifedipine rescued with 0.5-nM calyculin A. Number of replicates shown in corner. Scale bars, 500 μ m. (E) Live-cell orientation of GFP-labeled cell bodies in control and 200- μ M nifedipine-treated *ex vivo* cultures. Mean $\pm$ SEM is shown. (F) Brightfield and immunofluorescent maximum intensity projections of F-actin and nuclei in control and 500- μ M nifedipine-treated back skin explants. Control values shown in (C) were also used as the 22-h time point in (B). The 22-h values shown in (B) were performed under similar experimental conditions as the 12- and 32-h time points, but on a different day, so no statistical comparison is made, but the points are listed on the same graph for comparison. Scale bars, 50 μ m. * $p < 0.05$; **** $p < 0.0001$. Box plots represent 5–95 percentile. See also Figure S4.

L-type channel blocker, nifedipine (Atsuta et al., 2019), abolishes pattern formation *ex vivo* (Figure 5D). Additionally, inhibiting Ca^{2+} signaling interferes with cell alignment, consistent with a role for Ca^{2+} in increasing cellular contractility (Figure 5E). In turn, activating Ca^{2+} signaling through the addition of capsaicin appears to activate contractility (Figure S4).

To determine whether Ca^{2+} signaling facilitates pattern formation by mediating the necessary increase in contractility, we investigated whether ectopically increasing contractility in the absence of proper Ca^{2+} signaling restores pattern formation. Indeed, we can rescue the effect of nifedipine through additionally increasing contractility with calyculin A treatment (Figure 5D). In addition, we found that these same results can be achieved using the inhibitor of stretch-activated channels, GdCl_3 (Nakamura et al., 2001; Guilak et al., 1999), further implicating mechanosensitive channels in this process (Figure S4). Lastly, to test whether these observations *ex vivo* are indicative of an undescribed role for Ca^{2+} *in vivo*, we applied nifedipine to skin explants. Compared with the control, nifedipine abolishes the supracellular alignment observed at E6.5 and also prevents the formation of feather follicles (Figure 5F). Together, these results support a model where Ca^{2+} signaling is necessary for cells to tune contractility in response to changing extracellular mechanics. In doing so, Ca^{2+} signaling mediates an emergent cell-ECM co-evolution critical for generating an aligned supracellular structure and establishing the proper contractility and ECM conditions to transition to a pattern of aggregates.

Mesoscopic liquid phase properties enable spontaneous aggregate formation

We then investigated whether a regular pattern of aggregates is the predicted outcome of an aligned, contractile cell-ECM continuum. We therefore performed a theoretical analysis of how an aggregate pattern initiates. In identifying the general mechanism by which regularly spaced aggregates form, we may reveal how patterning is tuned by the biological and physical properties of the system. The simple geometry and the extent of experimental control render our *ex vivo* platform particularly suited for a theoretical analysis of the most salient features contributing to the initiation of an aggregate pattern. In particular, the orientation of the cells and ECM along the perimeter of the ring stipulates that the problem can be considered along a single dimension.

We therefore formalized our understanding of the effective material properties of the aligned cell-ECM layer into a continuum theory, relying on the following four assumptions.

(1) Our findings compel a description of the cell-ECM layer as a viscous material. In particular, we find that in our *ex vivo* assay, cells are enmeshed in ECM (Figure S5) and this cell-ECM layer undergoes irreversible rearrangements on the timescale of 1–10 h (Video S2), and our MPA results suggest that embryonic chicken skin possesses viscous fluid-like characteristics; namely, it continuously deforms, or flows, over time under constant applied force, and it retains that deformation after force is released (Figure S5). Studies of viscous properties in tissues

are underexplored, and viscosity is often excluded in favor of studies of elastic behavior (Cantini et al., 2020; Muller et al., 2013; Murray et al., 1983). (2) Because adherent cells exert contractile stresses on their environment (Schwarz et al., 2002; Schwarz and Safran, 2013; Tanimoto and Sano, 2014), we expect that dermal cells act as attractive elements when embedded in an ECM layer, drawing in adjacent cells. (3) We assume that the number and spacing of aggregates is determined at the early stages of the patterning process, i.e., the first 10–20 h. *In vivo*, the density patterns are read out at E7.5 (~12–24 h after patterning onset) by β -catenin signaling (Shyer et al., 2017), while *ex vivo*, patterns are stabilized on a similar timescale by the depletion of material in between aggregates. Thus, we aim to describe patterning onset and early aggregation and neglect effects such as spatial variations in rheological properties or dynamic parameter changes that may eventually arise from feedback between cells and the ECM upon strong aggregation. (4) Consistent with our observation that the timescale of patterning occurs on a much faster timescale than cell proliferation (Figure S5), we do not consider fluctuations in the total cell number.

Based on these points, we modeled the aligned cell-ECM layer as an active attractive fluid (Bois et al., 2011), obtained the hydrodynamic equations governing the dynamics of the cell density and material flow along the ring, and investigated the spontaneous appearance of patterns using linear stability analysis and simulations (Figures 6 and S6; Methods S1).

Our theory recovers that the self-attracting activity of contractile cells in the ECM is capable of generating the spontaneous formation of regular aggregate patterns (Figure 6A). Furthermore, our calculations reveal how two key parameters govern the capacity of the cell-ECM layer to form patterns: the cellular contractility and the hydrodynamic length. This length parameter characterizes the spatial range of flows in the material (Mayer et al., 2010) and depends on the mechanical properties of the ECM, such as the cross-linking between its components.

In particular, linear stability analysis and simulations yield a distinct relationship between these parameters and (1) the size and spacing of multicellular aggregates and (2) the timescale of pattern formation (Figures 6B, 6C, and S6; Methods S1). Specifically, we find that the number of aggregates that form should increase as contractile stresses generated by the cells increase, and patterning should take place within a shorter period of time. In turn, the model predicts that an experimental decrease of active cellular contractility would lead to fewer aggregates, which should form on a longer timescale. Moreover, an increase in the hydrodynamic length should lead to fewer aggregates and slower patterning.

In order to test the extent to which our theoretical calculations capture the biophysical dynamics of the cell-ECM layer, we compared our predictions with the experimental results. Our *ex vivo* platform, where we reconstitute multicellular dynamics in the most minimal context, allows for the perturbation of individual physical features and observations of the resulting pattern dynamics. It therefore serves as a multicellular diagnostic assay, providing a mechanical fingerprint for each population or perturbation. When we inhibit cellular contractility with the myosin inhibitor blebbistatin, the resulting pattern matches the predicted decrease in the number of aggregates. Conversely, treatment

with calyculin A, an amplifier of cellular contractility, increased the number of aggregates as expected (Figures 6B–6E). Our calculations also predict that the mechanical properties of the ECM influence cellular aggregation by setting the spatial range of cellular flows such that increasing viscosity, for example, leads to fewer aggregates and eventually stalls patterning (Figures 6A–6D). We pretreated collagen with different concentrations of ribose to modulate the cross-linking of collagen fibers and thereby alter the viscosity of the network (Stojkov et al., 2021; Tian et al., 2016). Indeed, at a concentration of 250 mM ribose, fewer aggregates emerge, and treatment with 500 mM ribose abolishes observable aggregation (Figures 6B–6E). Moreover, our experimental observations confirm the predicted changes in the patterning timescales for both manipulations (Figure S6). Further details of our analyses, the simulations, and the parameters used are provided in the Methods S1, the materials and methods, Table S2, and Figure S6. This theoretical study, with its experimental agreement, supports a scenario in which the emergence of an aligned supracellular structure sets the conditions for a contractile instability that resolves as the spontaneous emergence of follicles.

DISCUSSION

Mesenchymal mechanics can autonomously initiate periodic morphological pattern

Broadly speaking, morphological changes can involve an expansion of an already existing morphology (e.g., elongation [Mongera et al., 2018] and spreading [Morita et al., 2017]) or an increase in complexity, as is seen in the case of the emergence of follicles in the skin. *In vivo*, it has remained unresolved whether the initiation of follicle primordia occurs through the action of cellular components with different molecular profiles (Painter et al., 2018) or through a spontaneous self-organization of a homogeneous field of cells (Shyer et al., 2017). Although previous studies have presented *in vivo* and explant data consistent with both models, here we use a reconstitution approach to demonstrate a process of spontaneous cellular self-organization. Going forward, we anticipate that *ex vivo* reconstitution of morphological processes will be a critical step in de-coupling the effects of multicellular mechanics from molecular signaling.

This *ex vivo* system fills longstanding gaps in studies of collective cell behavior. In particular, we gain access to mesenchymal dynamics, whereas organoid and collective cell assays predominantly focus on epithelial cells and substitute mesenchyme with a commercial ECM. We also augment classic fibroblast culture on collagen (Stopak and Harris, 1982; Harris et al., 1984; da Rocha-Azevedo and Grinnell, 2013) by showing that through incorporating the proper cell density, ECM properties, and geometric constraints seen *in vivo*, mesenchymal self-organization is responsible for tissue-specific, regular patterns. A further advance is that we incorporate time-lapse imaging, providing a window into cell-ECM dynamics.

By reconstituting each of the critical steps observed *in vivo*, we provide evidence that these events are not the consequence of well-known signals from the epithelium or from forces provided from neighboring tissue layers. Rather, these events are autonomously generated by the dermis. Given the prevalence of

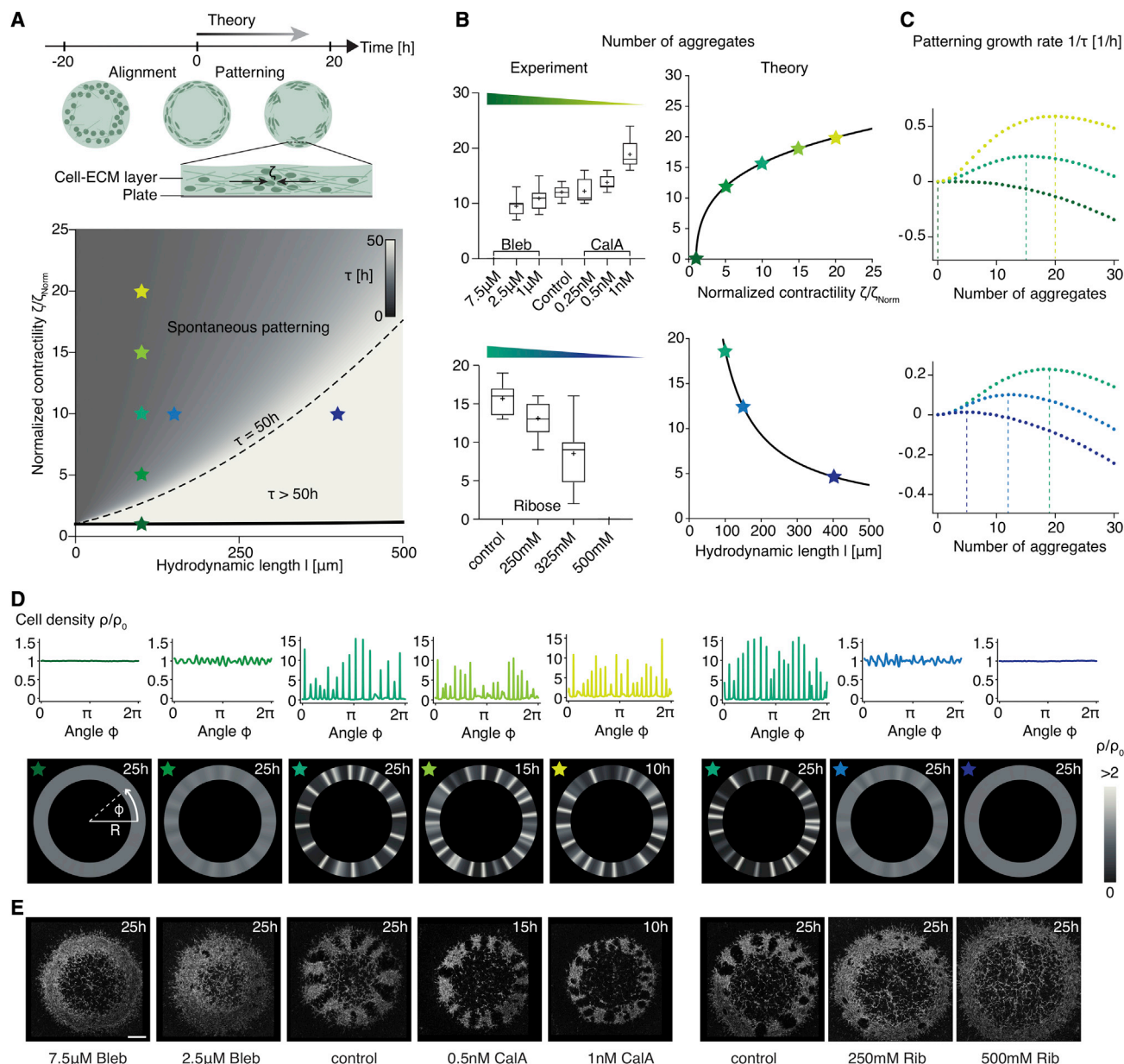


Figure 6. Theory and experiment show that collective contractility and ECM rheology modulate patterned multicellular aggregation

(A) (Top) Timeline of aggregate formation and schematic of the theoretical framework. We address the dynamics during the initial stages of patterning, after the cell-ECM system has aligned along the ring, approximately 20 h post plating, and model the ring as a one-dimensional active attractive fluid. (Bottom) Parameter diagram showing the timescale τ of the fastest-growing spontaneous patterning mode in dependence on the hydrodynamic length and the normalized contractility ($\zeta_{\text{norm}} = \gamma D$, with friction coefficient γ and effective diffusion coefficient D). Colored stars mark the parameter combinations used in the subsequent panels.

(B) Comparison between aggregate number variations measured in pharmacological perturbation experiments and predicted by linear stability analysis. We modulated myosin-driven contractility with blebbistatin (inhibition) and calyculin A (activation), and altered ECM cross-linking by pre-treatment with ribose. Box plots represent min-max values.

(C) Growth rates predicted by linear stability analysis for the dominant patterning modes.

(D) Snapshots of simulations using the different contractility and hydrodynamic length values marked in (B) and experimentally estimated parameters (Table S2). ρ_0 denotes the initial cell density along the ring.

(E) Transformed brightfield images of the patterning dynamics under pharmacological modulation of contractility and ECM cross-linking. Scale bars, 500 μm . See Figure S6 for a time point-by-time point comparison between simulations and experiments for these conditions, and the Methods S1 for details on the theory. See also Figures S5 and S6.

mesenchymal tissue layers across the embryo during morphological maturation, the sufficiency of cellular contractility as an intrinsic force generator may be conserved across tissues contexts. However, our results indicate that the mechanics of mesenchyme needs focused attention.

A consideration of the ECM has been underemphasized in studies of multicellular mechanics due to its diminished role in epithelial tissue models, in addition to the challenges that visualizing and manipulating the ECM poses. However, we find that the ECM is vital for setting the stage for a tissue-scale spontaneous transformation. Moreover, although mesenchymal collectives are intrinsically more disordered than epithelial collectives, we find that the alignment of the ECM serves to generate supracellular order.

Soft matter physical models as a conceptual framework for morphological transformation

In order to fully explain our data, we found it necessary to invoke concepts developed in the field of soft matter physics. First, we noted that supracellular order was actively created by nematic ordering of cells and the ECM. In the case of a developing tissue, such ordering can be advantageous for transitioning a disordered layer of mesenchyme into one that has long-range structural properties. Indeed, by probing the mechanical properties of the tissue, we find that nematic ordering accompanies a shift in material properties of the tissue. Such a transition is likely necessary as tissues generate stable morphological structures on the scale of hundreds to thousands of cells. Furthermore, this long-range order can serve to orient the individual behavior of cells—in this case, contractile forces—along a desired axis. Importantly, we note that such nematic ordering differs from ordering observed in passive systems because, in our biological case, the interactions between the individual units driving the alignment depend on active cytoskeletal processes within the units themselves.

While previous studies of morphological emergence have focused on elastic properties of tissues (Savin et al., 2011), here we found evidence of fluid-like properties as a fundamental driver of spontaneous multicellular self-organization. Rheological phenomena in development and disease have been previously noted but have yet to be applied to the study of non-trivial morphological changes in vertebrates driven by mesenchyme (Wang et al., 2020; Petridou et al., 2021). The instances where developing vertebrate tissues are modeled as fluids focus on flow in driving folding or spreading of a tissue or tissue region (Saadaoui et al., 2020). In contrast, our findings support a model whereby a contractile instability in an active attractive fluid spontaneously generates a morphological, periodic pattern. Analogous mechanisms for pattern formation occur in active fluids at other scales, including the clustering of myosin motors in the actin cortex (Hawkins et al., 2011; Bois et al., 2011; Alvarado et al., 2013; Hannezo et al., 2015; Nishikawa et al., 2017) and on cytotkinetic rings (Wollrab et al., 2016; Thiyagarajan et al., 2021).

Our results suggest that fluid instabilities, such as those observed when thin films of water on glass retract into droplets, offer a powerful organizing framework for periodic pattern formation in tissue biology.

Supracellular regulation bridges molecular events to tissue-level phenotype

The perspective that the fundamental mechanisms responsible for generating form are best framed on scales larger than the molecular scale was posited decades ago (Murray et al., 1983; Oster and Alberch, 1982). In contrast, in recent decades, supracellular mechanics have been seen as an elaboration of gene and molecular regulation leading to a focused search for gene regulatory events. Along these lines, studies addressing how a periodic pattern of follicles forms, focused predominantly on molecular events. The finding that tissue rearrangement and local activation of signals are dependent on forces generated by mesenchymal mechanics broadened the scope of investigation to include larger length-scales (Shyer et al., 2017). Here, we uncover how such force generation is initiated, uncovering regulatory events across biological scales.

Regulatory mechanisms respond to a change in system conditions and modulate it toward a particular outcome. In the embryo, regulation enables morphological robustness by channeling form toward a specific result rather than veering off course as a result of small fluctuations. Here, pivotal regulatory mechanisms were identified through an analysis of cell and ECM dynamics at the multicellular scale. Although the cell-ECM relationship is often depicted as static or one-directional (Levental et al., 2009; Fraley et al., 2015; Piotrowski-Daspiet et al., 2017), we find that the linkage between cells and the ECM evolves dynamically. Such a co-evolution regulates the relative levels of the two key biophysical parameters, contractility and ECM viscosity, assuring the formation of a particular pattern out of the entire phase space of pattern possibilities. The presence of stochasticity at the molecular and cellular scale during embryogenesis is commonly observed across organisms. Our results suggest that rather than combating stochasticity through additional molecular regulation, mechanisms at the multicellular scale exploit it to generate instabilities that resolve into new patterns of form.

The supracellular regulation observed here undeniably rests on the actions of molecular components, specifically in that the behavior of each cell within a dermal collective is determined by the genes expressed within it. Nevertheless, critical properties emerge at the supracellular scale that could not be discovered or understood through persistent dissection of molecular features alone. These results highlight a growing awareness that regulatory events occur at multiple scales and that these events can be simultaneous and linked. A major challenge in studies of morphogenesis will be to strategically select scales of study to find the proper foothold into multiscale processes. We propose that the multicellular scale can bridge the vast catalog of molecular states to higher order tissue structure in two ways. First, the multicellular scale can act as a phenomenological filter, allowing for more focused screening of molecular drivers given known cell properties of interest. Second, multicellular properties set the template for organ and tissue structure through spontaneous transitions that emerge from a supracellular continuum.

Limitations of the study

The mechanism we present focuses on the critical step of pattern initiation. However, this is just one key aspect of the

complex dynamic process of follicle formation. Once the follicle initiates in the skin, signaling molecules are activated through mechanotransduction. Future work will investigate how such signals modify the individual properties of skin progenitor cells, thereby altering the macroscopic properties of the dermal tissue to capture the initiating structure and then perpetuate its formation. Such studies of how signals shift dermal tissue properties will inform our theoretical account of follicle formation, as our current model does not account for how the pattern is cemented once initiated. Further, our study exploits the simplicity of symmetry breaking in just one dimension. Therefore, in future extensions of this mechanism, it will be necessary to consider how the emergence of the initial midline row mechanically primes the emergence of adjacent rows. Moreover, investigations of pattern emergence in two dimensions across a tissue field will be needed to address whether this mechanism explains follicle patterning during mammalian skin development.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.cell.2022.04.023>.

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AUTHOR CONTRIBUTIONS

A.E.S. and A.R.R. conceived of the project, designed experiments, and wrote the paper. K.H.P. designed and performed experiments and analyzed data. S.F.T. established the synthetic morphogenesis assay, performed experiments, and analyzed data. A.E. performed linear stability analysis and simulations of the theoretical model and gave technical support and conceptual advice. S.Y. assisted with assay development. F.L.E. initiated the synthetic morphogenesis assay. C.R.P. constructed the MPA platform and performed MPA experiments and analysis. All authors discussed the results and implications and commented on the manuscript at all stages.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Fibronectin	Abcam	Cat# ab6328; RRID: AB_305428
Collagen-I	Millipore Sigma	Cat #AB765P; RRID: AB_92259
Chemicals, peptides, and recombinant proteins		
Ribose	Millipore Sigma	R9629-100g
Blebbistatin	Abcam	Ab120425
Calyculin A	Sigma	C552-10μg
Nifedipine	Abcam	Ab120135
GdCl ₃	Tocris	4741
E-Capsaicin	Abcam	Ab141000
Critical commercial assays		
FLIPR Calcium 6	Molecular Devices	R8190
Deposited data		
Active Fluid Model of Mesenchyme Patterning	This paper	https://notebookarchive.org/2022-04-52goc4g
Experimental models: Cell lines		
DF1	ATCC	CRL-12203
Experimental models: Organisms/strains		
Wild-type Chicken Embryos	UConn Animal Science Dept.	White Leghorn
GFP Chicken Embryos	Clemson University	Roslin Green
Software and algorithms		
ImageJ	NIH	Version: 2.1.0/1.53c

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Dr. Amy Shyer (ashyer@rockefeller.edu)

Materials availability

This study did not generate new unique materials.

Data and code availability

All data reported in this paper will be shared by the [lead contact](#) upon request.

All original code has been added as a [supplemental information](#) and is available at Notebook Archive: <https://notebookarchive.org/2022-04-52goc4g>.

Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Cell lines

Primary cells were obtained from the dermis of embryonic day 8 back skins, and cultured in 1x DMEM (gibco) supplemented with 2% chick serum (gibco) and 10% FBS (gibco).

In vivo animal studies

Fertilized chicken eggs (white leghorn) were obtained from commercial sources, incubated at 37.5°C, and staged according to Hamburger and Hamilton. GFP-labeled embryos used in this study were obtained from Clemson University and maintained in a manner similar to wild-type embryos.

METHOD DETAILS

Embryos and dissections

Embryonic day 8 back skins, consisting of between 3–5 rows of emerging follicles, were dissected and the epidermis and dermis were dissociated by peeling apart the two layers after 15 minutes in cold calcium/magnesium-free HBSS. The extracted dermises from several embryos were combined and dissociated in a mixture of trypsin and collagenase.

GFP-labeled cells were isolated in the same manner, and mixed with wild-type cells at the appropriate concentration.

In vitro culture/ex vivo assay

After incubating in the trypsin/collagenase mixture for 10 minutes at 37°C, a single cell suspension was made by pipetting, and acellular tissue components were removed in a 40 μ m filter. The cells were diluted in full media (5% FBS, 2% CS) to between 1400 and 2100 cells/ μ L to achieve an appropriate annulus geometry. Cells were plated on collagen-I that was stored overnight at 4°C, and incubated prior to plating for 30 minutes at 37°C in a drying incubator. Cells were incubated at 37°C for 1 hour and then full media was added to the well. For all long-term imaging experiments, cells plated on collagen-I were incubated in a BioTek BioSpa 8 at 37°C with 5% CO₂ in a humid environment (~93% humidity) and were automatically transferred for imaging every hour.

Collagen plates were prepared by mixing type I collagen (rat tail high concentration, #354249, Corning) with DMEM and neutralized with NaOH to a final concentration of 1.0mg/mL (and a pH of roughly 7.35) and polymerized in 24-well plastic-bottomed plates (Corning) at a volume of 250 μ L for 30 minutes at 37°C.

For experiments where it was desirable to visualize the collagen matrix live, the collagen mixture was combined with 1.1 μ m spherical beads (Thermo Fisher) at a final concentration of 1:1000 prior to plating.

For glycation experiments, collagen gel was mixed with 0.5M ribose solution in 0.1% acetic acid to form glycated collagen gel solutions containing final concentrations of 0mM, 250mM, 325mM, 400mM, and 500mM ribose. Solutions were incubated at 4°C for 5 days, as described previously (Mason and Reinhart-King, 2013). Gels were subsequently prepared for ring culture by mixing with DMEM and neutralized with NaOH, as described above. For treatments to modulate cellular contractility *ex vivo*, blebbistatin (ab6328, Abcam) was added at 6 hours (Figure 4) or 19–20 hours (Figure 6) after plating and calyculin A (C552, Sigma) was added at 19–20 hours after plating. To abolish calcium signaling, nifedipine (ab120135, Abcam) was added at 6 hours after plating.

In experiments used to test the contractility of cells after different periods of time in our *ex vivo* assay, cells were removed from *in vitro* culture by incubating with 0.15% collagenase (type I, Worthington) for 5–10 minutes at 37°C. Collagenase was then removed by rinsing the cells in full media twice. Cells were then diluted to the appropriate concentration, and plated on 400 μ L of 1.5mg/ml collagen-I in a 24-well plate for a bead contractility assay, or on 250 μ L of 1.0mg/ml collagen-I for a *ex vivo* assay to measure contractility.

For the bead contractility assay, cells were plated in a 2 μ L drop of approximately 1700 cells/ μ L on collagen-I containing GFP beads. After plating, the cells and matrix were imaged every hour for the first 3 hours under a 4x objective to measure the collective contractile properties of the cells.

Micropipette Aspiration

Glass capillary tubes with 1.0 mm outer diam./0.75 mm inner diam. (TW100-3, World Precision Instruments) were pulled into micropipettes using a Sutter Instrument P-2000 laser-based micropipette puller and cut to various diameters using a Sutter Instrument ceramic tile for scoring glass (NC9569052, Fisher Scientific).

For each aspiration experiment, the sample was mounted in buffer between two glass coverslips and secured above the 10 $\times$ /0.30-NA objective of a Nikon ECLIPSE Ti2 inverted microscope, equipped with an Andor Zyla sCMOS camera. A BSA-coated micropipette was inserted between the coverslips and positioned flush against the sample region of interest; micropipette movements were controlled by Narishige's MN-4 coarse manipulator and MMO-4 micromanipulator. With the pipette abutting the region of interest, suction pressure was applied manually *via* syringe using a hydraulic system like the one described previously (Heinrich and Rawicz, 2005). This pressure was applied rapidly (<1 sec) to approximate instantaneity, and its magnitude was reported by a Validyne PS309 digital portable manometer.

Two different day 6.5 embryo back skin samples were aspirated at various sites in both the pattern-forming band of dermis ('stripe') and in the adjacent non-pattern-forming tissue ('non-stripe'); the regions were distinguishable by eye based on higher cell density in 'stripe'. A 62.8 $\pm$ 0.3 μ m-diam. pipette was used to aspirate $\geq$ 5 times per region. For each measurement, 7.9 $\pm$ 0.1 kPa of pressure was applied rapidly at $t = 0$ sec, held constant, and released rapidly at $t = 60$ s. Images were acquired every 95 msec from $t = 0$ sec to $t = 90$ sec.

Collagen-I gel measurements were taken similarly. Two different gel samples, prepared as described here, were each aspirated at $\geq$ 6 disparate sites into a 51.6 $\pm$ 0.2 μ m-diam. pipette under pressures of 8.2 $\pm$ 0.2 kPa. Pressure was applied at $t = 0$ sec, and then released at $t = 15$ sec, with images acquired every 75 msec for a total of 30 sec.

Image analysis was performed in ImageJ (Schneider et al., 2012). For each aspiration of each sample, extension of the sample into the micropipette was quantified over time; extension is the length L from the pipette aperture to the leading edge of the sample inside the pipette. When aspiration pressure is applied to a viscoelastic material like skin or collagen, extension increases rapidly (instantaneous strain) and then more slowly (creep strain). In a given time, a softer (more deformable) material will extend further into the

pipette than a stiffer (less deformable) one. Hence, extension L serves as a simple metric of mechanical deformability (Figure 2B; Figure S1). In Figures 2B and S1, respectively, L was measured 60 sec and 15 sec after aspiration pressure was applied.

Because extension increases over time, it reaches a maximum, $L_{\max}$, right before aspiration pressure is released, at which point it decreases to a relaxed value, L_{relaxed} . ‘Deformation recovery’ is calculated as $(L_{\max} - L_{\text{relaxed}})/L_{\max}$ (Figure 2B). If $(L_{\max} - L_{\text{relaxed}})/L_{\max} = 0$, the sample remains fully extended, without moving, after pressure is released; if $(L_{\max} - L_{\text{relaxed}})/L_{\max} = 1$, the sample completely recovers to its original shape (i.e. completely retracts from the pipette) after pressure is released.

In Figure 2B, *in vivo* data come from the ‘stripe’ region of the back skin, while *ex vivo* data come from the collagen-I gel. In Figure S1, data come from the ‘stripe’ region of the back skin, although the fluid-like characteristics observed in the ‘stripe’ region are also observed in the ‘non-stripe’ region. To obtain the average deformation curve plotted in Figure S7, we min-max normalized the deformation curve for each individual measurement (2 back skins, ≥ 4 measurements per skin), and then calculated the mean deformation across all measurements at every time point.

Immunohistochemistry and imaging

Ex vivo cultures were rinsed twice in PBS and fixed in 4% paraformaldehyde in PBS for 20 minutes. Skin samples were dissected from E6, E6.5, and E7 embryos, and then promptly fixed for 1 hr in 4% paraformaldehyde in PBS. After fixation, samples were rinsed in PBS, and PBS-Triton (PBSTr, 0.1% Triton X-100). *In vitro* cultures were blocked for 30 minutes and skin explants were blocked for 1 hr in 10% CAS Block in PBSTr (Invitrogen). All samples were stained in primary antibody overnight at 4°C. The following primary antibodies were used: fibronectin (1:200 *in vitro*, 1:100 *in vivo*, ab6328 Abcam), Col-I (1:100 *in vitro*, 1:50 *in vivo*, AB765P Millipore Sigma). Samples were subsequently rinsed in PBSTr, then incubated for 2 hours with the following secondary antibodies: Alexa Fluor 488 (1:300, Invitrogen), Alexa Fluor 555 (1:300, Invitrogen), Alexa Fluor 647 (1:300, Invitrogen). DAPI (1:1000, Invitrogen) was used to stain nuclei. ActinGreen 488 ReadyProbes Reagent (R37110, Molecular Probes), ActinRed 555 ReadyProbes Reagent (R37112, Molecular Probes), and Alexa Fluor 488 and 647 Phalloidin (Fisher Scientific, A12379, A22287) were used to stain actin. Immunofluorescent imaging of *in vitro* cultures was performed using a Nikon Eclipse Ti2 at 4x and 20x.

Calcium imaging was performed using the FLIPR Calcium 6 assay (Molecular Devices) according to manufacturer’s protocol. In brief, samples were rinsed twice with 1x HBSS, incubated in a 1:1 dilution of FLIPR Calcium 6 assay solution in HBSS for 2 hours, and imaged promptly. Images were recorded at 10x, under a 488nm laser every 10 seconds for a 30 minute period.

For live-imaging fibronectin, ab6328 was conjugated to Alexa Fluor 555 using the Alexa Fluor 555 Antibody Labeling Kit (A20187, Molecular Probes). Live-imaging of fibronectin, FLIPR Calcium 6 signaling, and multicellular dynamics under brightfield were performed using a BioTek Cytation 5 imaging reader, with the exception of the image in Figure 3B, which was imaged using an Incucyte Live-Cell imaging platform.

Fixed skin explants stained with antibodies were imaged using an LSM 880 with a 40x oil-immersion objective. Final images were a max intensity projection of 2-3 z-stacks. For decellularized tissue samples, decellularization was performed using 0.1% SDS for 30 minutes at room temperature, immediately followed by three rounds of rinsing in PBS. Skin explants imaged under brightfield to show gross follicle morphology over time were imaged using a Zeiss AxioZoom.V16.

Analysis and quantification

Ex vivo characteristics were quantified in ImageJ by drawing lines and outlining areas. Single values were obtained for each *ex vivo* assay, and were averaged over multiple samples. Wavelength was defined as the distance between the centers of adjacent aggregates (measured as circumference/#aggregates). Ring width was defined as the thickness of the annulus at the timepoint immediately prior to rip formation. Onset of aggregation was defined as the timepoint at which the first rip appeared *ex vivo*. Percent aggregation success was defined as the area of the annulus covered by cells at a final timepoint.

Directionality/Orientation of cells (actin/nuclei/cell bodies) and matrix fibers (FN) was quantified using the Directionality plug-in in FIJI. The Fourier components method was used on 600pi x 600pi cropped regions of *in vitro* cultures that were oriented with the annulus’ circumferential axis being parallel to 90°, or 300µm x 300µm cropped region of *in vivo*, *en face* wholemount regions with 90° being parallel to the midline. The measurements were sorted into 10 bins ranging from 0° to 90°. Parallel to the circumferential axis was defined as all measurements in the 80° and 90° bins.

Analysis of calcium signaling fluorescence intensity over time was calculated in ImageJ by randomly drawing an ROI over individual cells and then extracting the fluorescence values over time. Distance between adjacent peaks were determined using the SciPy peak-finding algorithm with thresholding false-peaks below the 90th percentile to reduce noise. Mean peak-to-peak values for a cell were reported. As the intensity was somewhat variable, background noise likely resulted in the incorrect calling of some peaks and the elimination of some true peaks.

PIV measurements were made using the Iterative basic PIV plug-in in ImageJ. An initial interrogation window size of 128 pixels was applied with a correlation threshold of 0.60. The vector magnitude values were obtained and averaged over two trials.

The experiments were designed such that there would be both replicates representing data from separate experiments, and technical replicates within experiments. In our *ex vivo* system, data were excluded or not quantified if morphology was disrupted by technical plating issues.

QUANTIFICATION AND STATISTICAL ANALYSIS

Directionality measurements were plotted using matplotlib in Python. All other measurements and statistical analysis were performed in Prism 8. In line plots measuring cell directionality over time, error bars are SEM. In radial plots, error bars are SD. Box plots show min-max distribution. When calculated, p-values were determined using a Student's t-test with Welch's correction. For details on image analysis and quantification, see the [method details](#) section and the statistical parameters list ([Table S1](#)).

Model parameters used in the simulation (below) are shown ([Table S2](#)). Briefly, ring radius was calculated in ImageJ by drawing a line from the center of the ring to the midpoint between the outer and inner edge of the annulus. This was performed from the center to three different regions of the annulus and the lengths were averaged. The initial density was calculated by counting the number of cells in a region of the annulus and dividing this number by the arclength over that region (at the point between the inner and outer edge of the annulus). The diffusion coefficient D was calculated as $D = \langle r^2 \rangle / (4t)$, with $t = 1$ h. The squared displacement r^2 was calculated by tracing the initial and final x,y-coordinate (two-dimensional) for each cell in the ImageJ plug-in, Trackmate ([Tinevez et al., 2017](#)), and averaging was done over 73 cells.

In addition, for comparison of density profiles between simulated and experimental samples, density was quantified in experimental samples from phase contrast images of rings at five timepoints by drawing a line over the entire annulus and getting the intensity profile in ImageJ. These intensities were then background subtracted by the region depleted of cells and normalized to the average intensity along the ring immediately prior to patterning (I_0).

Simulations

Calculations, simulations and plotting for [Figures 6](#) and [S6](#) were performed with Wolfram Mathematica v.12.1 and v.12.3. The original code is provided as a [supplemental information](#). Details of the linear stability analysis are given in the [Methods S1](#).

Equations 1 and 3 (provided in [Methods S1](#)) were numerically solved with Mathematica's NDSolve function using periodic boundary conditions and a near-uniform initial condition for the density given by

$$\rho(s, t_0) = \rho_0(1 + \mathfrak{N}(s))$$

with $\mathfrak{N}(s)$ a third-order polynomial interpolation function between 1000 pseudorandom numbers drawn from the interval $[-0.1, 0.1]$, equally spaced along the ring perimeter.

Model parameters were estimated experimentally wherever possible ([Table S2](#)). Contractility and hydrodynamic length parameters for simulating the perturbation experiments shown in [Figures 6](#) and [S6](#) were chosen by comparing the number of aggregates predicted by linear stability analysis to the measured values ([Figure 6B](#)), and using previously reported values reported for the hydrodynamic length in actomyosin-driven supracellular flows as an orientation ([Behrndt et al., 2012](#); [Saha et al., 2016](#)).

On long timescales, aggregate coarsening occurs as reported previously ([Bois et al., 2011](#)). Timepoints shown in [Figure 6D](#) were chosen to capture patterns prior to aggregate coarsening.

Supplemental figures

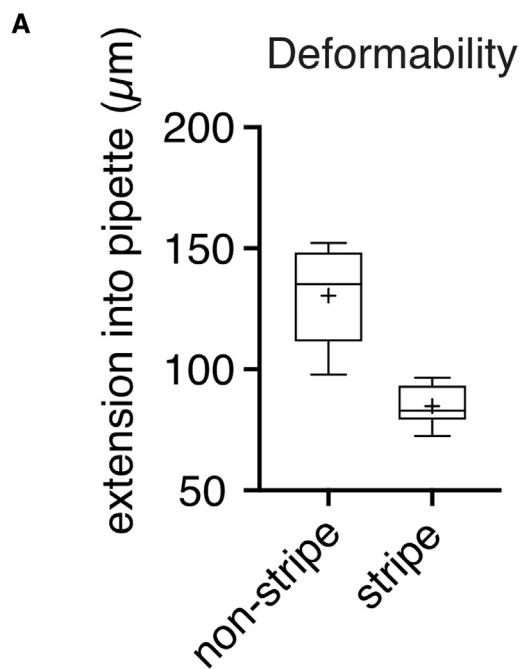


Figure S1. Cell and ECM alignment changes tissue material properties, related to Figure 1

Deformability of non-stripe and stripe at day 6.5 by extension into pipette (60- μm diameter) in μm using micropipette aspiration.

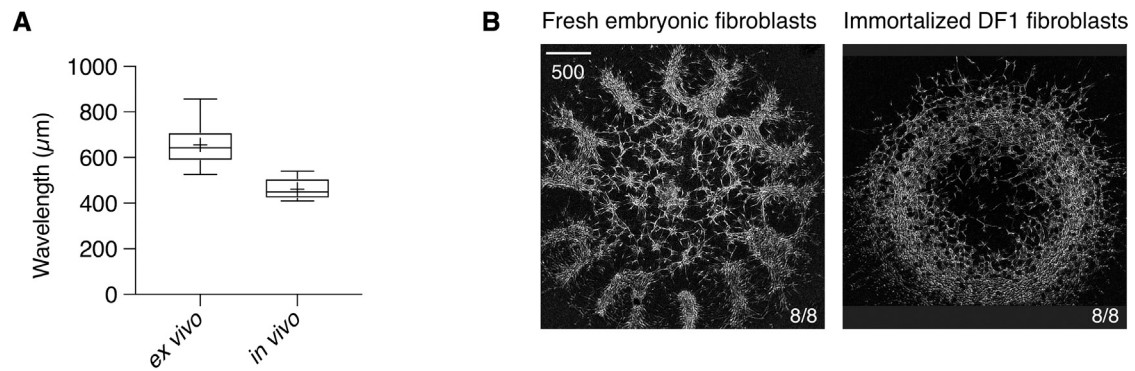


Figure S2. Behaviors intrinsic to primary dermal fibroblasts generate patterns *ex vivo* with a similar scale to *in vivo*, related to Figure 2
 (A) Wavelengths, the distance between adjacent aggregates (*ex vivo*) and nascent follicles (*in vivo*), were measured at 48-h *ex vivo* and at E7.0–E7.5 *in vivo*.
 (B) Transformed brightfield images of fresh embryonic dermal fibroblasts and immortalized DF1 cells plated *ex vivo*. Number of replicates over two trials shown in corner. Scale bars in μm .

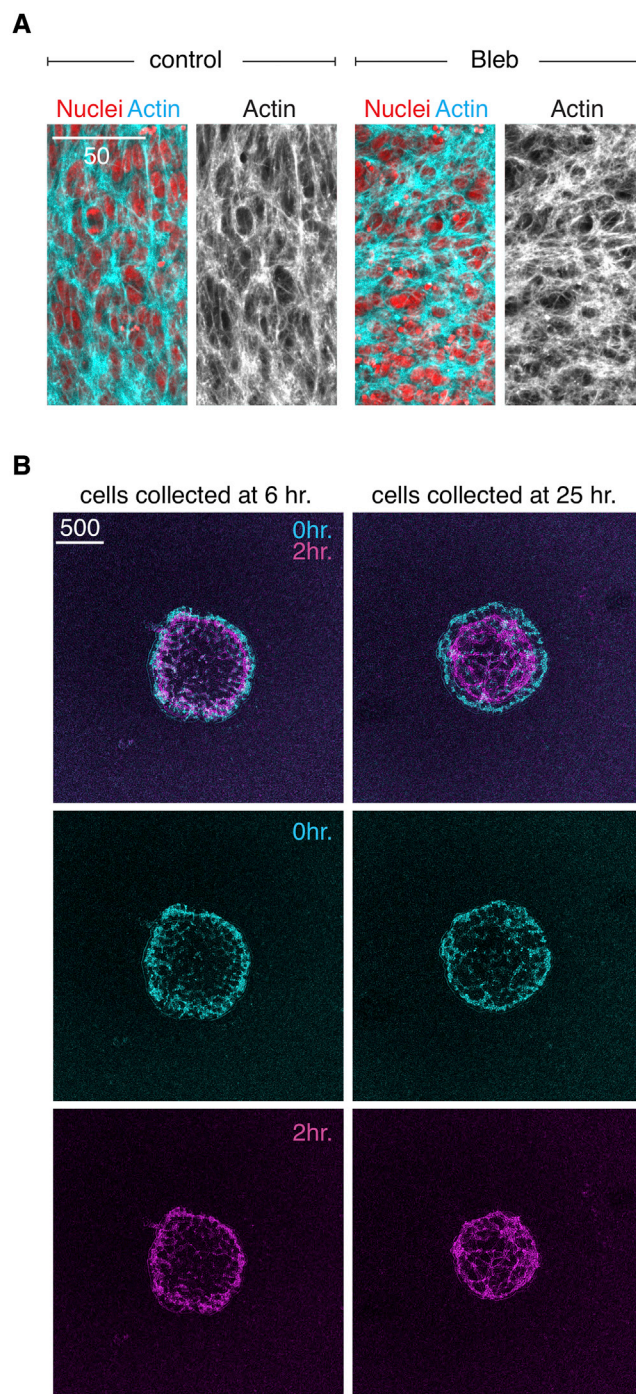


Figure S3. Contractility is necessary for supracellular alignment *in vivo* and also increases over time in *ex vivo* cultures, related to Figure 4
 (A) Maximum intensity projections of control and 50- μ M blebbistatin-treated explants stained for nuclei and F-actin. The same control is used in Figure 5F, because these two experiments were done simultaneously with the same DMSO control. Scale bars, 50 μ m.
 (B) Replated samples isolated at 6 or 25 h from the *ex vivo* assay for collective TFM are shown. It is notable that no apparent difference in cell density or geometry exists at the 0-h time point, imaged immediately following replating (cyan), but that the samples replated from 25 h show more contraction of the gel by 2 h. Scale bars, 500 μ m.

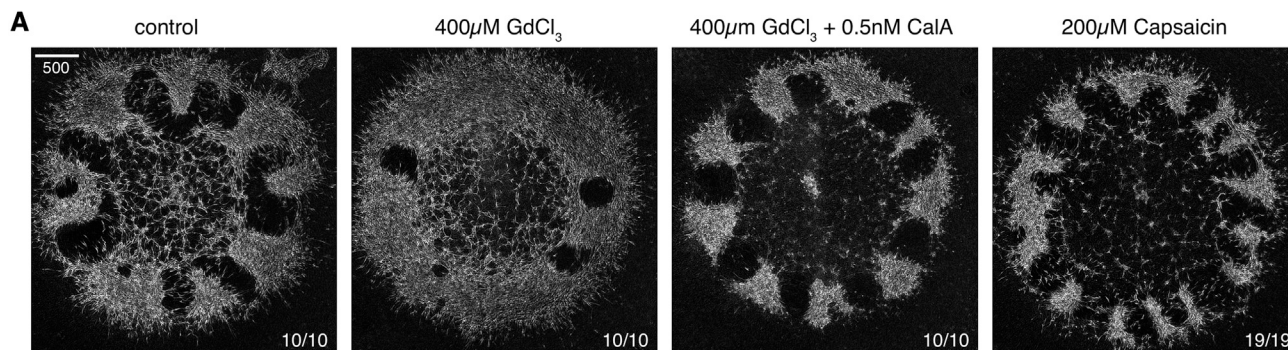


Figure S4. Calcium signaling regulates patterning by tuning contractility, related to Figure 5

(A) Transformed brightfield images of control, GdCl₃, GdCl₃ + calyculin A, and capsaicin-treated *ex vivo* cultures. Number of replicates over two trials shown in corner. Scale bars in μ m.

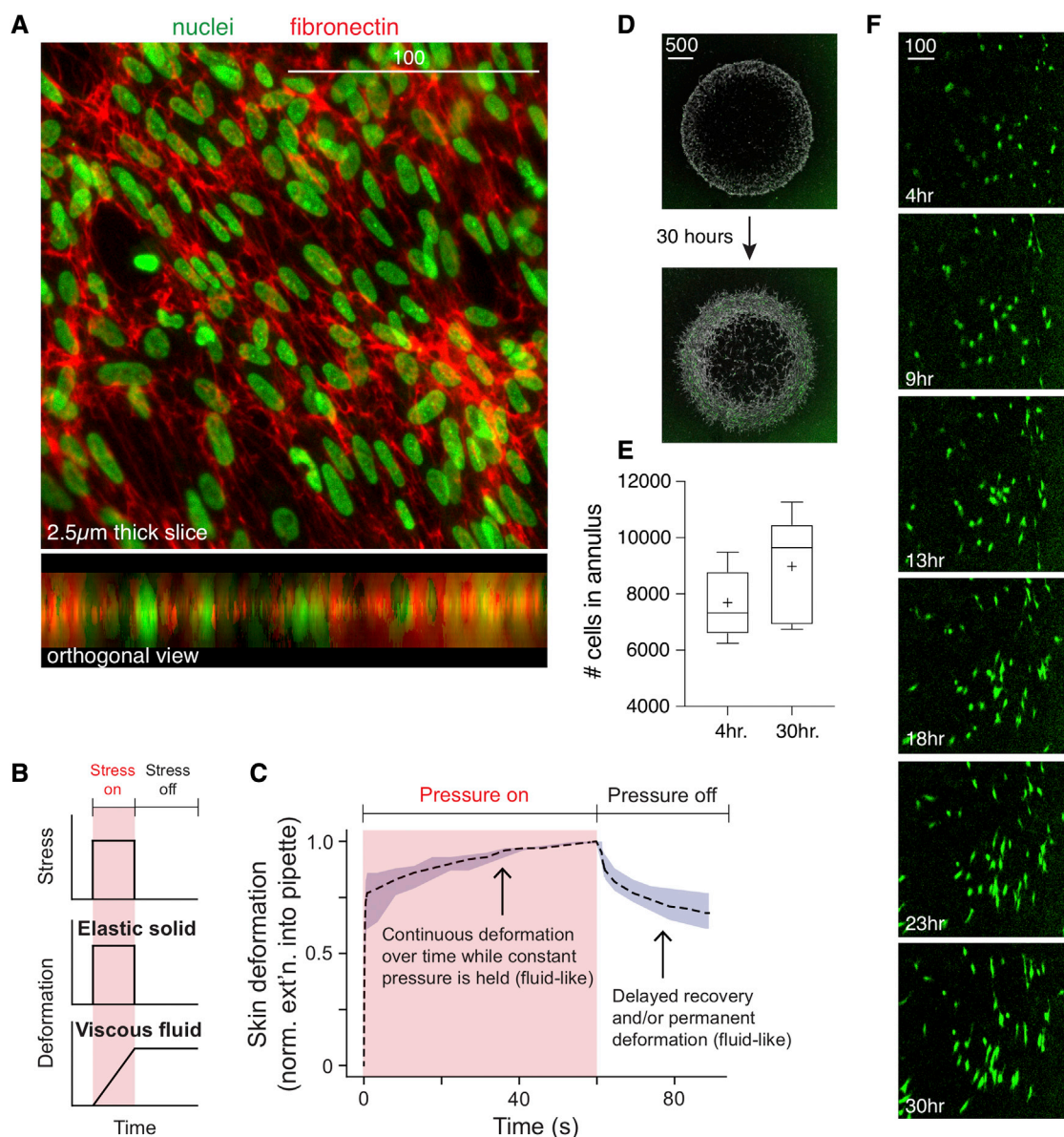


Figure S5. Dermal cells and ECM form a layer *ex vivo*, and display fluid-like behaviors *in vivo*, related to Figure 6

(A) Image of a single 2.5- μ m slice showing spatial overlap between nuclei (green) and fibronectin (red) in the z direction. A 3D projection of the entire z stack was compiled and an orthogonal view was recreated to show the overlap between nuclei and fibronectin in the z direction. Scale bars, 100 μ m.

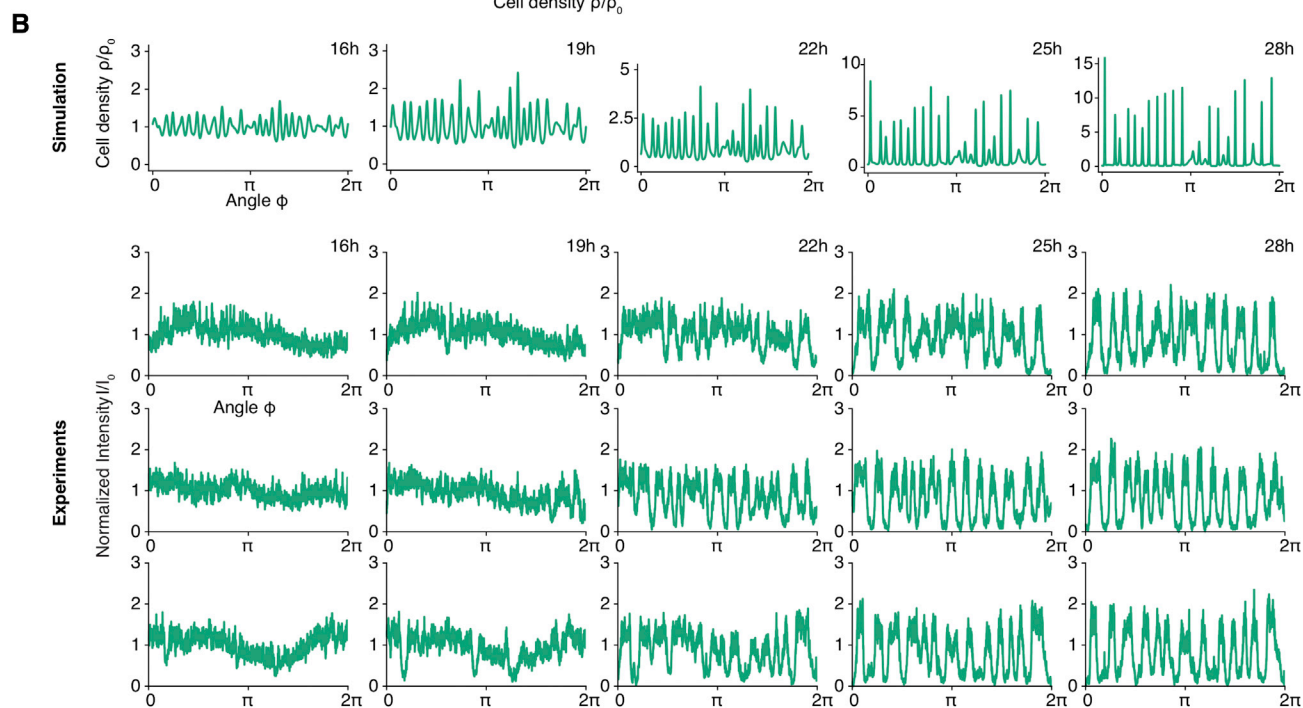
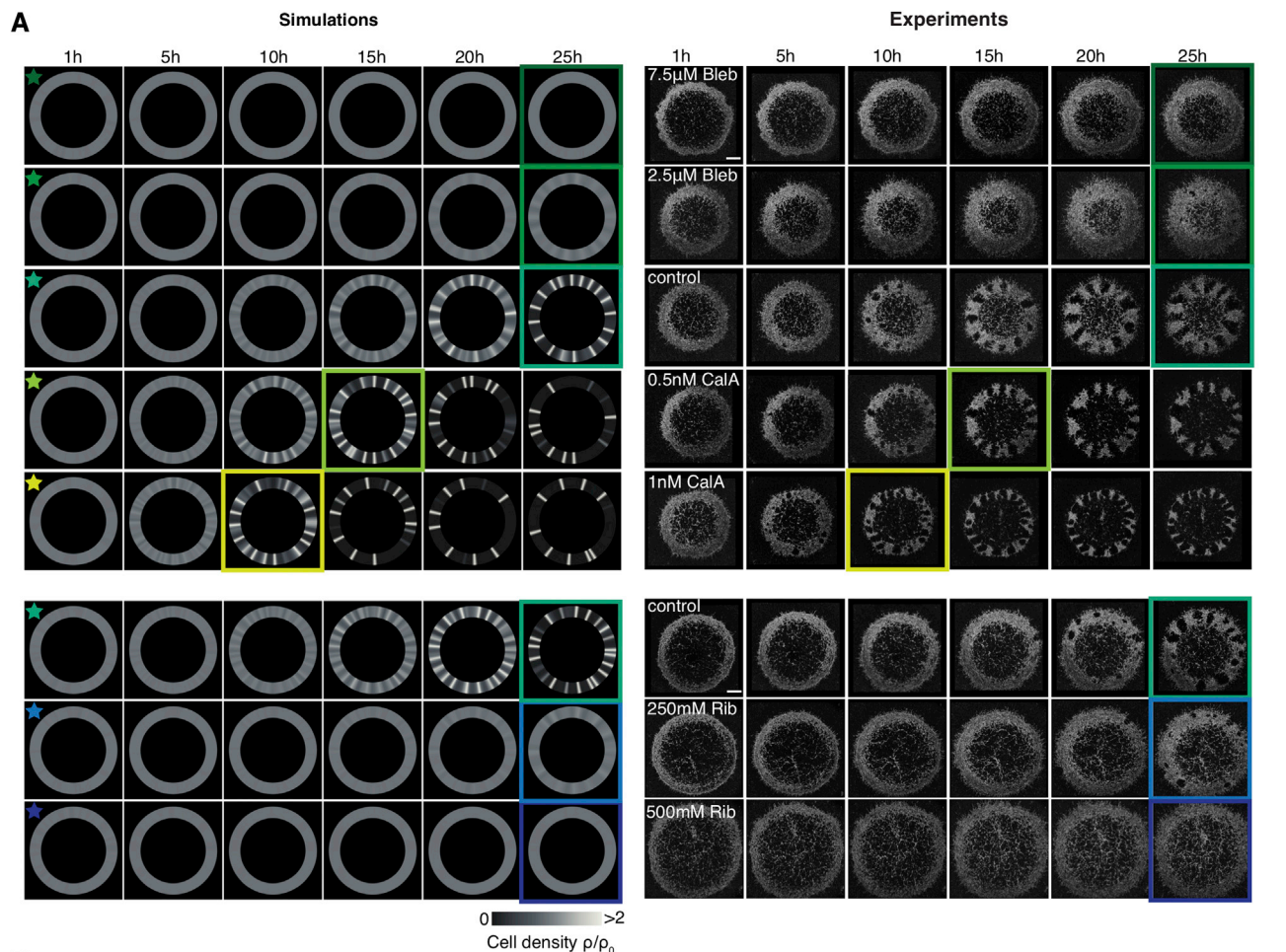
(B) Schematic diagrams: in response to a constant applied stress (top), an ideal elastic solid deforms instantaneously and then ceases to move; when the stress is turned off, the elastic solid instantaneously recovers its original shape (middle). By contrast, an ideal viscous fluid continuously deforms under constant stress, never coming to rest, and then retains that deformation after stress is released (bottom).

(C) Day 6.5 embryo back skin was pulled under constant pressure into a 60- μ m-diameter pipette. Black dashed line shows skin deformation over time, averaged across all measurements; blue shading indicates minimum and maximum measurements. Two back skins, ≥ 4 measurements per skin.

(D) Low-magnification view of *ex vivo* assay with brightfield cells and sparsely labeled (5% GFP) cells at 4 and 30 h after plating.

(E) High-magnification images of sparsely labeled (5% GFP) cells.

(F) Quantification of the number of cells at 4 and 30 h. Boxplot with minimum to maximum is shown, with plus sign showing the mean. Scale bars in μ m.



(legend on next page)

Figure S6. Comparison between simulated and measured patterning dynamics, related to Figure 6

(A) (Left) Simulations of the active fluid model using the different contractility and hydrodynamic length values marked in Figure 6 and experimentally estimated parameters (Table S2). Panels shown in Figure 6 are highlighted with frames. (Right) Transformed brightfield images of the patterning dynamics under pharmacological modulation of contractility and ECM cross-linking. Scale bars, 500 μm . Timestamps on simulations and experiments follow the convention shown in Figure 6A, with $t = 0$ corresponding to the aligned uniform state at 20 h post plating. For large contractilities, the theory predicts an eventual decrease in the number of aggregates due to coarsening, as reported previously (Bois et al., 2011). This is not observed in the experiments, possibly due to an overall depletion of material between the aggregates.

(B) Comparison between the simulated dynamics of the density profile and three measured examples from time points after the onset of patterning for control conditions. Note that the simple linear relationship between cell density and contractile stress used in the model (see Methods S1) predicts very large density peaks. These are not observed in the experiments, suggesting that the magnitude of active stress is limited at large cell densities. Additionally, pixel intensity saturation in the experimental images prevents estimation of the maximum density value.